

Basic concepts: numbers, operations, non-existence and indetermination

Author: Cecilia Alvares

1) Concept of numbers and mathematical operations

The most fundamental thing that should precede anything when starting to get into calculus is the idea of **mathematical operations** and the existence of **numbers**. The philosophical question on whether mathematics was invented or discovered by us lies out of the scope here, and so does the historical aspect of it. Numbers could have found their utility in our lives from the simple need of describing the quantity of a given object you have – books, shoes, dresses, coins. Basic mathematical operations such as sum, subtraction, multiplication and division, also have definitions that are also very natural to us. On the other hand, basic aside, there are other mathematical operations whose origin is less straightforward to imagine or grasp. Examples are exponential, logarithm, sine, cosine, and so on. Regardless of being basic or not, it is important to realise that all mathematical operations have an underlying well-established definition:

Any numerical operation can be regarded as a command-line or a pre-defined algorithm¹ that tells what to do with an input to get an output. They are, each, like a recipe to be followed, a tie that links an input and an output in a way to generate a numerical pattern (or mapping).

¹ Algorithm = sequence of commands to execute a given task.

A **numerical pattern or mapping** is defined here as the correspondence between different numbers, used as input, and their result from performing the mathematical operation.

Practical example: summing 2 any input number generates a given specific output so that ultimately the operation implies a specific mapping between the set of inputs and the set of outputs.

Additionally, just like each individual operation implies in a mapping of its own given by its definition, **coupling mathematical operations sequentially also yields a specific mapping of its own**. Given the huge amount of existing operations and the infinite amount of combinations there can be, an infinite amount of mappings can be created.

To think: independently of the debate on whether mathematics was fundamentally invented or discovered, it is interesting to note that the mathematical operations as we know them (i.e. association of name and command line) are rather something that we defined. Human race could have defined - and still can define - a set of other operations that aren't still there.

Other than defining mathematical operations, it is important to define a mathematical language in order to make statements. Writing it down is useful in order to help with a reasoning or even communication with others. In this context, specific symbols were defined for the different mathematical operations. A convention was also defined for the order according to which the operations must be carried out. In the latter case, usage of parenthesis has also been incorporated and used to specify the desired ordering. Here are some definitions important in mathematical language:

Equations and **inequations** are **mathematical statements**. Equations are used to state an equality between the two sides of a the statement, while inequations are used to express an inequality between the two sides (one side is larger, smaller or different than the other). A set of operations arranged in mathematical language can be referred to as **(mathematical) expression**, regardless of whether they amount to a statement (i.e. an expression containing an equality or inequality symbol).

Below are some examples of equations and inequations, alongside with whether or not the statement is true or false:

$2 + 5 = 7$	TRUE!
$2 + 2 = 5$	FALSE!
$2 + 2 \neq 7$	TRUE!
$6 + 2 * 4 = 14$	TRUE!
$(6 + 2) * 4 = 14$	FALSE!
$7 < 10$	TRUE!
$2 + 2 > 5$	FALSE!
$(8 + 2) * 2 > 30$	FALSE!

Although the examples above are simple, we note that assertions can also involve more sophisticated operations, such as limits, derivatives and integral, which we cover in further chapters.

2) The concept of indetermination vs non-existence: undefined outputs

In the context of the definition of mathematical operations, **understanding the definition of indetermination and non-existence is crucial:**

Cases of indetermination and non-existence occur when it is not possible to determine the result of mathematical expression. In the case of indetermination, it is not possible **because more than one output is possible for a given input** in the scope of a given mathematical operation involved in the expression. In the case of non-existence, it is not possible **because there is not a single possible output for a given input** in the scope of a given mathematical operation involved in the expression.

I would like to distinguish here indetermination from indeterminate forms, as the latter is formally defined only in the context of a mathematical operation called “limit” (of a function), which we will see later on. The word “indetermination” may be retarded in this study as an informal definition made for any mathematical operation (i.e. no outputs) to distinguish from scenarios where not a single output is possible.

Perhaps the best practical example to illustrate how indetermination and non-existence can arise within a mathematical expression as well as the difference between them lies within division. Division is **defined** within mathematics to be the inverse of multiplication – in other words, **it is its fundamental definition, with the algorithm of division being thus intrinsically tied to that of multiplication.**² Therefore, fundamentally, any time that a division operation is carried out, the reasoning that allows to reach the result should formally pass through the reasoning underlying the definition of multiplication. With that said, here goes a practical example:

Practical example: ask yourself the result of $7/0$. By definition, the question that should lead to the result is “which number you can multiply 0 by and obtain 7 as a result?”. Is there such a number? Is it possible to think of one possible number that can be the answer to that? The answer is no. This brings to the table the definition of **non-existence: there is no possible result**. On the other hand, ask yourself what is the result of $0/0$, which translated into reasoning, leads to the question “what is the number we can multiply 0 by and obtain 0 as a result?”. I bet you can find many numbers that answer that question: in fact, there are not only two or three possible results, but an infinite amount as any actual number can be a valid answer. This brings us to the

² Credits: Flaviana Dutra, mathematician at PUC-Minas, Belo Horizonte, Brazil.

definition of **indetermination**: there is more than one possible result.

While indetermination forms and non-existence are conceptually different, in both cases there is no answer (or output) for the given input: the output in both cases is said to be **undefined**.